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Extraction and classification of risk-related sentences from securities reports

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ABSTRACT

With the drastically changing business environment, it is difficult even for experts to properly extract and classify risk statements from securities reports, which contain large volumes and unstructured information. Several methods have been proposed, but the existing methods face difficulty in dealing with flexible risk expressions. This study presents an open-domain risk-analysis framework that combine the strengths of both humans and machines. They include defining appropriate business risks and constructing supervised data based on those definitions. Risks were then extracted and classified from the securities reports of a representative group of Japanese companies. We confirmed the limitations of pattern matching, and the usefulness of contextual analysis methods. We also confirmed the importance of constructing supervised data based on appropriate guidelines of data classification. This study presents a framework that quickly and effectively derives the risk structure of a given industry or company from vast and unstructured information.

Introduction

As the environment in which companies operate changes drastically, it becomes more important to assess corporate risks quickly and appropriately. According to Truant et al. (2017), recent policy changes in sustainability reporting have expanded the disclosure of ethical, social, and environmental risks in financial and socio-environmental reporting, thereby emphasizing the importance of risk identification (Truant et al., 2017). The impact of the environment, economy, and society on businesses is a risk factor that most companies cannot ignore. It has been highlighted that not engaging in activities that contribute to achieving the sustainable development goals (SDGs) and environmental, social, and corporate governance can itself be a risk (METI, 2019; Godfrey et al., 2009). Moreover, investors who evaluate corporate value emphasize the content of business continuity plans developed in response to the COVID-19 pandemic (Mukherjee et al., 2020).

Corporate non-financial information, such as risk, is an important source of information, which is difficult to quantify. In assessing corporate value, companies' non-financial information attracts greater attention. Conventionally, various methods have been proposed for calculating corporate value using indicators such as actual traded stock prices, future free cash flow, and the values listed on the balance sheet (Brigham et al., 1985). However, it is essential to check non-financial information because it is not possible to infer content such as performance factors from financial information alone (Christensen et al., 2016) Furthermore, securities reports, which listed companies are required to issue, comprise not only financial statements but also a wide range of items, including performance factors and potential companies' risks. In Japan, disclosing risk in securities reports is obligatory.

However, it is impossible for a person to extract, classify, and analyze all the necessary texts, which are vast and diverse. Moreover, governments and companies are becoming increasingly active in disclosing risk information (METI, 2017). With an increase in the number of risk descriptions, as mentioned above, the amount of information requiring processing has grown exponentially (Cohen et al., 2012; Dyer et al., 2017; Symes et al., 2017). The amount of information that humanity produces is increasing every year, making it impossible for even experts in the field to grasp it in its entirety (Sasaki et al., 2020). The actual evaluation of a company's value requires the confirmation of multiple securities reports, including those from previous years and other companies in the same industry.

In this context, there is a growing need for efficient information processing using methods such as text mining in the field of artificial intelligence. Bao & Datta (2014) and Dyer et al. (2017) provided a method for analyzing companies' textual risk disclosures to discover risk factors (Bao & Datta, 2014; Dyer et al., 2017). Huang et al. (2020) and Chang et al. (2021) assessed corporate value evaluation using text mining, and their results confirmed that there was an improvement

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in the accuracy of value assessments for companies making investment recommendations (Chang et al., 2021; Huang et al., 2020). Kurian et al. (2020) used machine learning and keyword analysis to analyze incidents and attempted to mitigate the risk of oil sands operations (Kurian et al., 2020). Concerning SDGs, Sebestyén et al. (2020) converted over 100 national reviews on SDGs published by different countries into a network of keywords using text mining algorithms (Sebestyén et al., 2020). Kumar et al. (2021) argued that risk analysis using text mining and NLP is effective for management in terms of a better understanding of user needs (Kumar et al., 2021).

Challenges exist in these machine approaches. The number of keywords used is generally limited, which restricts the amount of information that can be expressed for each document during text mining, thereby making it difficult to obtain new knowledge. Not all cases have established data for risk analysis in advance such as the study involving the German credit data set by Pandey et al. (2021) (Pandey et al., 2021). Li (2010) pointed out that unstructured textual information in financial disclosure documents is often more difficult to understand with conventional computer programs because of irregularities and ambiguities compared to data stored in the field format in databases or annotated in documents (Li, 2010; (EDINET - Electronic Disclosure for Investors' Network 2022). Human analysts' understanding exhibits a higher degree of accuracy than that of computer programs. From this perspective, Huang and Li's study is significant because researchers performed risk classification manually. In their study, researchers subjectively identified 25 risk factors after reviewing annual reports (Huang & Li, 2011; Financial Service Agency 2018). To some extent, this approach addressed the abovementioned problem of needing domain knowledge. However, there exist a wide range of risk factors, covering all of which is not possible. Lewis and Young noted that as reports from several companies increase, the shapes of documents become more diverse, and the lack of standardization and structuring makes analysis more difficult (Lewis & Young, 2019; Fu et al., 2020).

Using the computer-based approach alone incurs some problems, but so does using the human-analysts approach alone; therefore, an integrated approach is important. This perspective has not been adequately studied in the context of RM. It is important to discuss this perspective in a digital society where machines and humans are supposed to be emphasized.

When analyzing risk, it is important to correctly extract the text of risk factors (Campbell et al., 2014; Godfrey et al., 2009). Additionally, the clustering of risk factors should be performed because identifying industry- or period-specific risks in practical applications is crucial (Masson & Montariol, 2021; Hegde & Rokseth, 2020). To implement appropriate extraction and classification for large-scale data, it is necessary to construct appropriate correct-answer data. However, this must be evaluated objectively. Furthermore, it is necessary to be able to consider the context, which is not sufficient for traditional keyword-based PM.

The challenge of this study is to overcome the ambiguous representation of financial documents. This research's academic contribution concerning analyzing risks from unstructured financial documents can be summarized through the following two points: First, it proposes a framework that utilizes the advantages of both computers and humans to extract and classify risks that cannot be handled using conventional PM. Second, it demonstrates that an assessable risk definition is essential for this purpose. Although the construction of supervisory data is necessary to leverage the power of supervised machine learning, there has been no mention of the evaluation of supervisory data in the context of overall business risk.

Extracting and classifying useful risk statements from ambiguous or unstructured information is very costly, whether using humans or computers. It is a time-consuming task for experts, and the same is true for the construction of teacher data for the computer. With risk factors changing daily, conventional pattern matching does not allow for flexible context-aware risk analysis. Limited research has discussed the importance of defining risk and evaluating teacher data, which should be positioned upstream in the analysis of risk. This study presents a framework for a series of risk analyses: proper definition, construction of teacher data, risk extraction, risk classification, and risk assessment of firms and industries. It is shown that by defining risk appropriately, a model can be constructed that does not require a high level of expertise and that a realistic number of teacher data can be used to immediately contribute to risk analysis practice. This will allow each company to construct, as needed, teacher data appropriate to its era at low cost.

The contribution of this research to practice is to provide a framework that enables companies to conduct flexible risk analysis in accordance with the times, as the risk environment surrounding companies undergoes significant changes.

Related work

More and more companies are looking to adopt big data analytics and decision support systems to help managers make data-driven decisions by taking advantage of hidden trends and signals in the data (Arjun et al., 2021; Kushwaha et al., 2021; Unhelkar et al., 2022; Kushwaha et al., 2021; Arjun et al., 2021; Unhelkar et al., 2022). And it is also becoming possible to extract risks from bibliographic information using machine learning. Some studies have investigated the risks described in contextual annual reports and financial statements (Fu et al., 2020; Wei et al., 2019). If risks can be extracted, they can be fully utilized as corporate assessments (Aven & Krohn, 2014; Creedy, 2011; Paltrinieri et al., 2019), making this a worthwhile issue to address in any industry.

Many risk analyses are conducted in closed domains; Carracedo et al. (2020) investigated the impact of the COVID-19 pandemic on companies through text mining analysis using the statistical software R (Carracedo et al., 2020). Similarly, Zhao et al. (2019) aimed to identify risk factors in the oil market using text mining techniques (Zhao et al., 2019). Hegde & Rokseth (2020) conducted a keyword search and retrieved relevant from Scopus and Engineering Village databases articles and used machine learning methods to present a structured review of publications to assist in engineering risk assessment (Hegde & Rokseth, 2020). Countries facing similar problems and adopting similar development strategies were then clustered and analyzed. These studies focused on specific events or markets and analyzed risks; Koff et al. (2021) investigated how COVID-19 affected the job market, but limited their discussion to the scope of public sector job platforms and risks (Koch et al., 2021). Others have discussed specific risks and impacts - for example, risks in the insurance sector - from the perspective of using text mining for risks surrounding companies (Rawat et al., 2021). These closed-domain risk analyses are highly specific because they allow for the assessment of risks specific to that domain. While effective in each domain, without domain knowledge, proper assessment is difficult.

Since there are unique patterns in each closed domain, it is necessary to extract keywords that provide clues to the risks. One of the most used methods for extracting risk involves the use of keywords. For example, Kurian et al. (2020) proposed a method for building an efficient incident reporting system using machine learning and keywords (Kurian et al., 2020). This method, called pattern matching (PM), requires the collection of many keywords that can be used as clues in advance. For this purpose, the knowledge of domain-specific experts is also indispensable. Therefore, conventional research, as typified by PM, cannot properly extract and classify keywords in the absence of clues due to a lack of domain knowledge. It would be difficult to extract risks related to things that have not happened before using only keywords of existing risk factors. In the current situation, where there are even more diverse risk factors, it is important to consider the context to properly classify risks.

In this study, we deal with domain-independent risks and attempt to analyze risks in domains that are open in terms of their definition and classification. The goal is that through open domain risk analysis, teacher data can be extracted with a minimum of knowledge and ap-

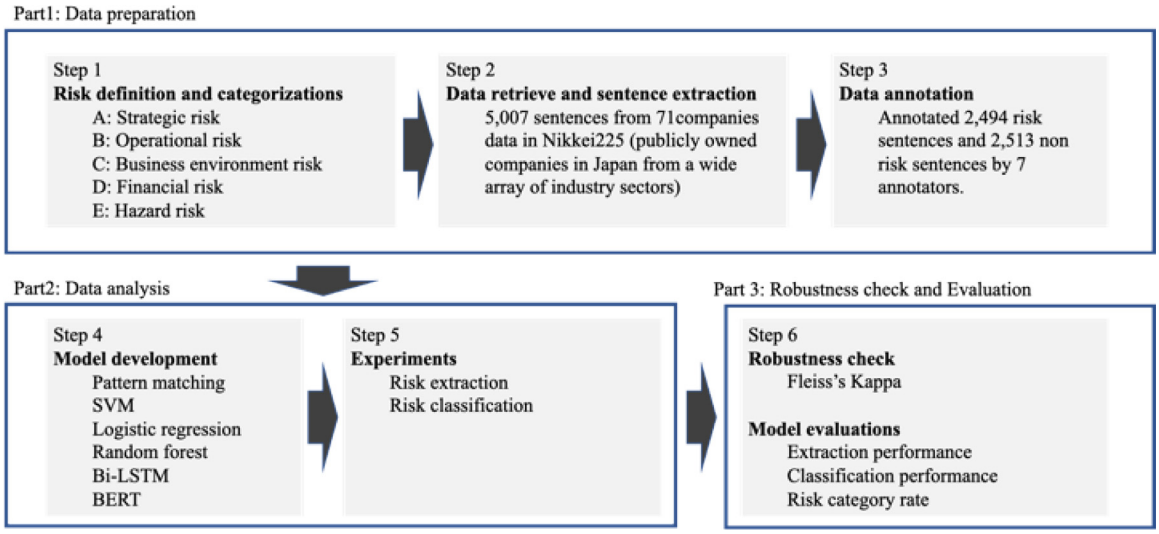

Fig. 1. An overview of methods.

plied to generic industries. For this purpose, an integrated approach is important, since a computer-based approach alone is problematic, but a human-based approach alone is also problematic. This perspective has not been adequately considered in the context of risk management. It is important to discuss this perspective in a digital society, where the emphasis should be on machines and humans.

Methods

Overview

The methodology used in this study comprised three major parts: data preparation, data analysis, and robustness checks and evaluation. Additionally, these parts comprise six steps. An overview of this method is shown in Fig. 1. The following subsection provides an outline. In Section 3.2, we describe each step in more detail.

Data preparation

The first step is to define and categorize the risk (Step 1 in Fig. 1). Before extracting and categorizing risks, it is necessary to clarify the definition of risk. Additionally, based on the results of several previous studies, risks are classified into five risks. The definition and categorization of risks are described in detail in Section 3.2.

Next, data collection was conducted (Step 2 in Fig. 1). In this study, the target is securities reports, which are disclosure documents stipulated by the Financial Instruments and Exchange Act. Therefore, it had an almost uniform format. The target companies for the extraction of securities reports were representative companies in the Japanese stock market. This allowed us to extract a well-balanced group of companies.

The next step was to annotate the data (Step 3 in Fig. 1). Seven annotators annotated the data based on the categorization defined in Step 1. We attempted to ensure the reliability of training data by adopting annotations that were agreed upon by multiple people. The details are described in Section 3.4. As an objective criterion for evaluating label results, we used the coefficient which measures the degree of agreement between evaluations by multiple evaluators (Section 3.6).

Data analysis

The next step was to build the model (Step 4 in Fig. 1). We used PM, machine learning (support vector machines [SVMs]), logistic regression (LR), random forest (RF) algorithm, and deep learning (bidirectional long-short-term memory [bi-LSTM] and BERT) as analysis methods in this study. An overview of these methods is presented in Section 3.5 Model Development.

After building these models, we conducted risk sentence extraction and classification experiments using the multiple models built in Step 4 with the sentences annotated in Step 3 as training data (Step 5 in Fig. 1).

Robustness check and evaluation

Finally, to evaluate the experiments, the extraction and classification performances were calculated and verified. Simultaneously, as a concrete case, we will contribute to the discussion with a discussion on a specific industry (Step 6 in Fig. 1).

Risk definition and categorization

When extracting risks, it is important to clearly define them. Risk has been defined in various ways in different fields. For example, the literature states that "The labeling rule is to label a sentence as risk-sentence if it includes the notion of uncertainty, and if at least one other element from the Risk triplet is present."[23]. However, the purpose of this study is to use the obtained information for corporate value evaluation; hence, a more appropriate definition is needed. The international risk management standard, ISO 31,000: 2018 Risk Management – Guidelines, defines risk as "the effect of uncertainty on objectives" (International Organization for Standardization, 2018). The 2019 revision of the Cabinet Office Ordinance on the Disclosure of Corporate Information, which specifies the contents of securities reports, states that the risks to be described are "matters that may have a significant impact on the judgment of investors, such as unusual changes in the operating results and other conditions of consolidated companies, dependence on specific customers, products, technologies, etc., unique legal regulations, trading practices, management policies, the occurrence of important litigation cases, etc., and important matters concerning directors, major shareholders, affiliates, etc." (International Conference on Harmonisation Q9, 2005).

This study defines risk as "the uncertainty of future outcomes that can be obtained by taking actions." Here, "outcome" refers to a company's performance, and uncertainty refers to a situation in which we know what state or outcome will appear but not its probability of appearing; that is, a situation that falls under "ambiguity under uncertainty" in the classification of uncertainty based on the decision-making environment (Brashers & Hogan, 2013).

For this study, the risks that were extracted based on the definitions described in the previous section were classified into several cat-

egories to facilitate the understanding of the types of risks that are frequently described in the text. According to Casualty Actuarial Society Enterprise Risk Management Committee. (2003), business risks can be classified into "strategic, operational, financial, and hazard risks." (Casualty Actuarial Society Enterprise Risk Management Committee, 2003).

However, according to Kaplan & Mikes (2012), there are internal and external risk perspectives; operational risk and financial risk can be considered internal risks, while hazard risk can be considered an external risk (Kaplan & Mikes, 2012). It is difficult to determine whether this the case for strategic risk. Therefore, in this study, we reclassify strategic risks, especially those caused by external factors, such as "business environment risks." Therefore, this study defines five categories of company risk: strategic risk, operational risk, business environment risk, financial risk, and hazard risk. Each of these risks is described as follows:

A company voluntarily assumes some risk in obtaining superior returns from its strategy (Casualty Actuarial Society Enterprise Risk Management Committee, 2003). In other words, strategic risk most affects an organization's ability to implement its strategy and achieve its business goals (Kaplan & Mikes, 2012). Examples of strategic risks include capital investment, changing needs, human skills' outflow, reputational damage, competition, customer demands, demographic, social and cultural trends, technological innovation, capital availability, and regulatory and political developments.

Operational risk arises within an organization and is related to the day-to-day problems that the organization faces in achieving its strategic goals (Kaplan & Mikes, 2012; Lewis & Young, 2019). Operational risk is caused by people, processes, and controls, such as business operations, empowerment, information, and business reporting (Casualty Actuarial Society Enterprise Risk Management Committee, 2003; Financial Service Agency, 2018).

Some risks arise from events outside the company and are beyond its influence and control (Li, 2010). Strategic risks include perfection, customer preference, technological innovation, and regulatory and political obstacles. Among these, risks arising from economic and social trends, such as changes in the competitive environment and demographic changes, are the most significant (Financial Service Agency, 2018; Li, 2008). These risks have been reclassified as business environment risks.

Financial risks include price (asset values, interest rates, exchange rates, commodities, etc.), liquidity (cash flows, call risk, opportunity cost, etc.), credit (defaults, downgrades, etc.), inflation or purchasing power, and hedging basis risk (Casualty Actuarial Society Enterprise Risk Management Committee, 2003). These are the effects that financial assets and liabilities are subject to due to market principles, such as interest rates, exchange rates, commodity prices, liquidity risks, and credit risks (Financial Service Agency, 2018; Li, 2008).

Hazard risk arises from exposure to property, liability, or human loss (Financial Service Agency, 2018). For example, fire and other property damage, wind, flood, other natural disasters, theft and other crimes, personal injury, and business interruption. Disease and disability (including work-related injuries and illnesses] and liability.

Table 1 shows the risk classification and specific examples of each classification.

Data retrieval and sentence extraction

The subject of this study is the extraction and classification of risk based on Japanese securities reports. EDINET (Electronic Disclosure for Investors' Network) is an electronic disclosure system for documents based on Japan's Financial Instruments and Exchange Law (44). We attempted to randomly extract 5000 sentences corresponding to the business risk item from the securities reports that the major companies comprising the Nikkei Stock Average disclose as digital data on EDINET. These sentences were used for the analysis Table 2.

Data annotation

Based on the obtained sentences, we created training data. The first step in creating training data is to annotate sentences with binary values to determine whether they are risky or non-risky sentences. Next, the extracted sentences were annotated using risk classification. In other words, two annotation levels were used to build the training data. Annotation was performed by seven individuals—five workers and two supervisors—familiar with securities reports. Seven Japanese people in their 20 s were selected, without bias, from the author's department to annotate and supervise. Belonging to the field of business administration, they are experienced in reading Japanese securities reports. We confirmed that they possessed sufficient knowledge to annotate and conduct text mining.

The five annotators, assigned 1000 sentences each, judged whether each sentence is a risk sentence based on the criteria shown in Section 3.2 Risk definition and categorization. At the same time, they labeled the sentences that were judged to be risk sentences in terms of the risk category they fall under. For every 5000 sentences completed by the five annotators, supervisor A performed the same process for 5000 sentences. Supervisor B processed 5000 sentences similarly. The decisions of the five workers and two supervisors were independent, and they did not exchange information with each other. If two or more of the three answers (annotator's label, supervisor A, and supervisor B) assigned the same label to a certain sentence, it was treated as the final label. The evaluation of the annotation results is described in detail in the robustness check section.

Model development

In this section, the classification methods used in this study are introduced. The classification methods are PM, SVMs, LR, RF algorithm, bi-LSTM, and BERT. SVM, LR, and RF are representative methods that are often discussed as evaluation targets for text classification methods (Pranckevičius & Marcinkevičius, 2017). In all the methods described below, morphological analysis by MeCab was used to extract the morphemes used as features during learning. Moreover, CaboCha was used as an entailment analyzer (Kudo & Matsumoto, 2002).

(1) Pattern matching (PM) by word form

PM is a method of defining string patterns in advance and checking whether the patterns appear in the target sentence. We used the accuracy of PM in extraction and classification by keywords as a benchmark to compare other methods.

String patterns are defined using regular expression. In this study, we classify the patterns of risk expressions in securities reports into the following two categories: As mentioned previously, word-string patterns are expressed as regular expressions. The first pattern comprises conjunctions used in logical sentences. The subordinate clause of a compound sentence is the former, and the main clause is the latter. The compound sentence expresses a relationship (causality) in which the truth of the proposition in the former part is related to the truth in the latter part. These sentences satisfy the definition of risk when the former part has not yet occurred; if the former part has already occurred, they do not satisfy the definition. Additionally, words that can be replaced by the above conjunctions are treated in the same way. The second type of expression describes the content of the event (proposition) as much as possible or can be replaced by such an expression. Even if these conditions are met, the proposition is excluded if it represents the outcome of the corporate activity, such as "the business performance will decline" or "the financial position will deteriorate." This is because the aforementioned propositions correspond to "outcomes" in the definition of risk, which is different from the content to be extracted (the reason for extracting is that the outcome is uncertain). The first and second types of expressions above are referred to as hypothetical and possible expressions, respectively.

Table 1
Specific examples of risk classification.

Risk classification	Definition	Risk description	A Concrete example	Refs.
A Strategic risk	A risk that arises in management judgment and decision-making in the execution of strategies such as business investment and marketing. This risk can be controlled by the company and is caused by top management actions.	Management strategy	Capital investment, existing business reform	Casualty Actuarial Society Enterprise Risk Management Committee (2003); Financial Service Agency, (2018); Kaplan & Mikes (2012); Lewis & Young, 2019; Li (2010), (2008)
		Marketing	Changing needs, technological innovation	
		Personnel system	Human resource outflow	
B Operational risk	A risk that major business activities will not function because of the failure of IT systems or facilities, strikes, etc. This risk cannot be controlled by the company and is caused by the actions of middle management in the field.	Products and Services	Product defects, recalls	Casualty Actuarial Society Enterprise Risk Management Committee (2003); Financial Service Agency 2018; Kaplan & Mikes (2012); Lewis & Young (2019); Li (2008)
		Legal and Ethics	Patent disputes, internal fraud	
		Environment	Environmental pollution, waste disposal	
C Business environment risk	A risk arising from changes in the competitive (business) environment, such as changes in economic conditions or the strengthening of legal regulations. This risk cannot be controlled independently by the company and is caused by the standard actions of people and companies.	Politics	Revision of laws, political instability	Casualty Actuarial Society Enterprise Risk Management Committee (2003); Financial Service Agency (2018); Kaplan & Mikes (2012); Lewis & Young, (2019); Li (2008)
		Economy	Economic fluctuations, rising raw materials prices	
		Society	Rumors, consumer movement	
D Financial risk	A risk related to finances, such as fund procurement and exchange rate fluctuations. This risk cannot be controlled by the company (i.e., the risk can be reduced by hedging transactions, but it cannot be eliminated; the company cannot interfere with the market itself).	Asset management	Stock price fluctuations, loan losses	Casualty Actuarial Society Enterprise Risk Management Committee (2003); Financial Service Agency (2018); Kaplan & Mikes (2012); Lewis & Young, 2019; Li (2008)
		Settlement	Interest rate fluctuations, exchange rate fluctuations	
		Capital and liabilities	Increase in funding costs	
E Hazard risk	A risk of disasters such as earthquakes and typhoons, war, civil strife, and crimes such as theft. This risk cannot be controlled by the company; it cannot interfere with the event itself, though it can mitigate the risk through measures such as insurance. The risk is caused by factors that cannot be interfered with artificially or by the unusual actions of people, companies, or nations (war, terrorism, and crime).	Natural disasters	Earthquakes, adverse weather conditions	Casualty Actuarial Society Enterprise Risk Management Committee (2003); Financial Service Agency (2018); Kaplan & Mikes (2012); Li (2008)
		Accidents	Fires, equipment failure	

Table 2
Keywords in risk sentences.

Risk classification	Keywords
Strategic risk	development, innovation, merger, competition, research, advertising, customers, employment, human resources, new products, capital investment, advertising, strategy, alliances, needs, acquisitions, duds, advantage, turnover, new business
Operational risk	violation, concealment, contamination, defect, environment, management, defect, infringement, strike, product liability, lawsuit, harassment, quality, fraud, deficiency, conflict, mistake, executive, recall, leakage, labor dispute
Business environment risk	image, reform, supply, business climate, economy, soaring prices, supply and demand, demand, consumer movement, credit, tax, tax system, system, commerce, rumor, boycott, trade, law, legislation, mass media, media
Financial risk	credit rating, bad debts, stock price, exchange rate, finance, interest rate, financial planning, assets, land price, derivatives, bankruptcy, real estate price, bad debts, securities
Hazard risk	virus, fire, infectious disease, weather, malfunction, accident, natural disaster, information system, flood, war, power failure, terrorism, weather, theft, civil war, windstorm, riot

The main objective of our study is to enable early risk management by extracting risks without using existing keywords. However, it is necessary to verify the performance of risk classification methods using keywords before evaluating new methods. Therefore, as in the determination of risk expressions, besides supervised machine learning and deep learning, we manually determined the words and phrases that could be regarded as particularly expressive of the content of the risks described in the securities reports.

(2) Support vector machines (SVMs)

Another model is the SVM, a method of constructing a two-class pattern discriminator using the simplest model of a neuron, the linear threshold element. In this study, the SVM was adopted as an effective method for binary classification because the training data were not large.

From a set of training samples, the parameters of the linear threshold element are learned using the criterion of "margin maximization." SVMs have supervised machine learning models in which a discriminative hyperplane is constructed in the data space, and only the most confusing data points (i.e., most likely to be misjudged under noisy conditions) are selected and trained, such that the hyperplane is as distant from them as possible. A set of hyperplanes can be constructed in high-dimensional space to perform classification and regression. The larger the margin, the smaller the generalization error of the classifier. Thus, proper separation was achieved by the hyperplane with the largest distance from the closest training data point of a given class. A binary output value was calculated for the input feature vector using a discriminant function (linear discriminant function).

In the implementation, SVM-light was used as a learner for risk sentence judgment, and scikit-learn and OneVsRestClassifier were used for risk classification (Pedregosa et al., 2011).

(3) Logistic regression (LR)

LR is a multivariate analysis method used to predict and explain the dependent variable in qualitative data. As a method that can be analyzed without being significantly affected by the scale and distribution of the data, it is expected to have a certain effect even if the dataset for risk statement classification is complex. It explains and predicts the probability of a single binary variable using multiple explanatory variables. Generally, machine learning algorithms make assumptions regarding the shape of the decision boundary. In the case of LR, the decision boundary is assumed to be linear. LR is known to be difficult to overfit because the decision boundary is linear and not complex. The LR can also be used to examine the magnitude of the effect on the target variable. It is a multivariate analysis method used to predict and explain the dependent variables in qualitative data. Scikit-learn was used for implementation, and OneVsRestClassifier was used for risk classification (Pedregosa et al., 2011).

(4) Random forest (RF) algorithm

The RF algorithm is a method in which several decision trees are created by random sampling, thus allowing duplication, and the final prediction value is determined by taking a majority vote of the prediction results from each tree. This algorithm is a simple learning method; however, it can identify and predict better than a typical decision tree. Since RF samples the input field, it is expected to be fast and accurate, even for datasets with numerous features. In this study, we adopted this method because it is expected to be effective when the number of features used for risk classification is large.

Another useful feature of the RF algorithm is that it does not suffer from overfitting when the forest size increases. The implementation is based on scikit-learn, and OneVsRestClassifier is used for risk classification (Pedregosa et al., 2011).

(5) Bi-LSTM

Bi-LSTM has demonstrated high performance in several natural language processing tasks (Chen et al., 2016). We used long short-term memory (LSTM)—a type of recurrent neural network developed to handle serial data (Hochreiter & Schmidhuber, 1997). In LSTM, we used bidirectional information from the beginning to the end of a sentence, and vice versa; thus, we implemented bi-LSTM. This is because there are cases where not only the forward but also the backward context is necessary, for example, word disambiguation; bi-LSTM often performs more effectively. Particularly because securities reports have long sentences, it is necessary to consider both the forward and backward contexts; we considered the possibility that information may be lost if we consider only one direction. We used PyTorch to implement the bi-LSTM. For risk classification, we used the same learning method as that used for risk sentence judgment for each of the five risk categories (five iterations of binary classification learning).

(6) BERT

BERT is a large-scale language model based on pretraining proposed by Devlin et al. (2018), and it has been trained effectively in various tasks related to natural language processing (Devlin et al., 2018; Jiao et al., 2019). In this study, we used it to extract risk sentences through fine-tuning. The model was trained on the Japanese Wikipedia published by the Inui Laboratory at Tohoku University. To use BERT in this experiment, we used three methods and a combination of their results. The methods comprised BERT_(a) ordinary multilabel classification, BERT_(b) simultaneous risk sentence extraction and classification learning (six multilabel classifications in total), and BERT_(c) binary classification learning for each of the five risk classifications.

We adopt BERT_(b) because the risk sentences that fit into each risk category comprised approximately 10% of the total sentences. We expected that the performance would improve by applying binary classification, particularly for models that need to handle large amounts of data, such as BERT. Fig. 2 shows a diagram of the BERT model used in this study. Fig. 3 shows a graph of the risk classification using binary classification.

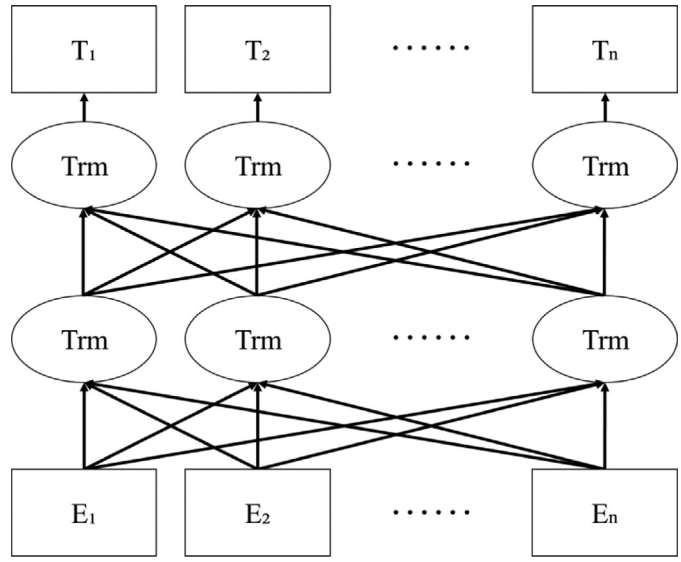

Fig. 2. BERT model (Devlin et al., 2018).

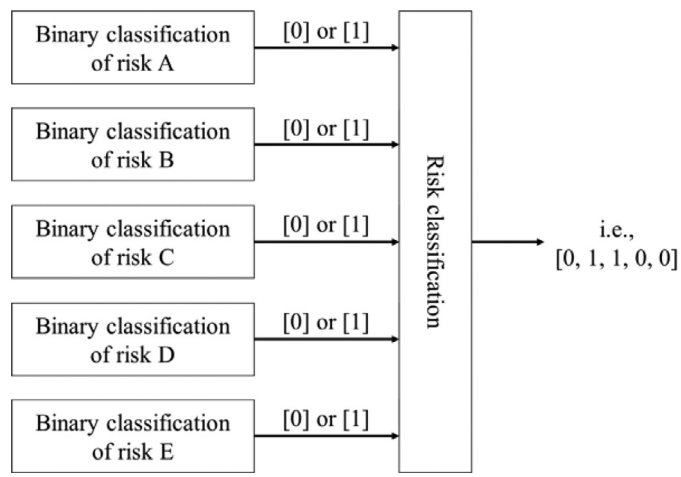

Fig. 3. Multilabel classification based on binary classification.

BERT_(c) was adopted because we predicted that the total number of risk sentences containing each risk category would vary; therefore, training them separately would produce a more accurate outcome because variations in the number of labels in the learning process may lead to low accuracy in a few labels, even when the overall accuracy is high.

In this study, the same importance was assigned to each risk classification. However, when trying to align the quantity of each dataset, there is a risk of missing necessary information when reducing data. Additionally, when increasing the amount of minority data, there is a risk of overlearning owing to data padding. Therefore, the accuracy of each risk category (especially those with a smaller quantity than the others) can be improved by eliminating imbalance regarding the number of positive labels (the number of risk statements that contain that risk category) by learning them separately.

These methods were combined because accuracy can be improved by combining the results of correcting imbalances and overlearning, which may be a problem with the data used in this project for the reasons mentioned above. BERT_(a) and BERT_(c), which look similar, were combined because, in an experiment using only sentences that were judged to be risky, the performance of financial risk (risk D) and hazard risk (risk E) in risk classification was higher when method (a) was used,

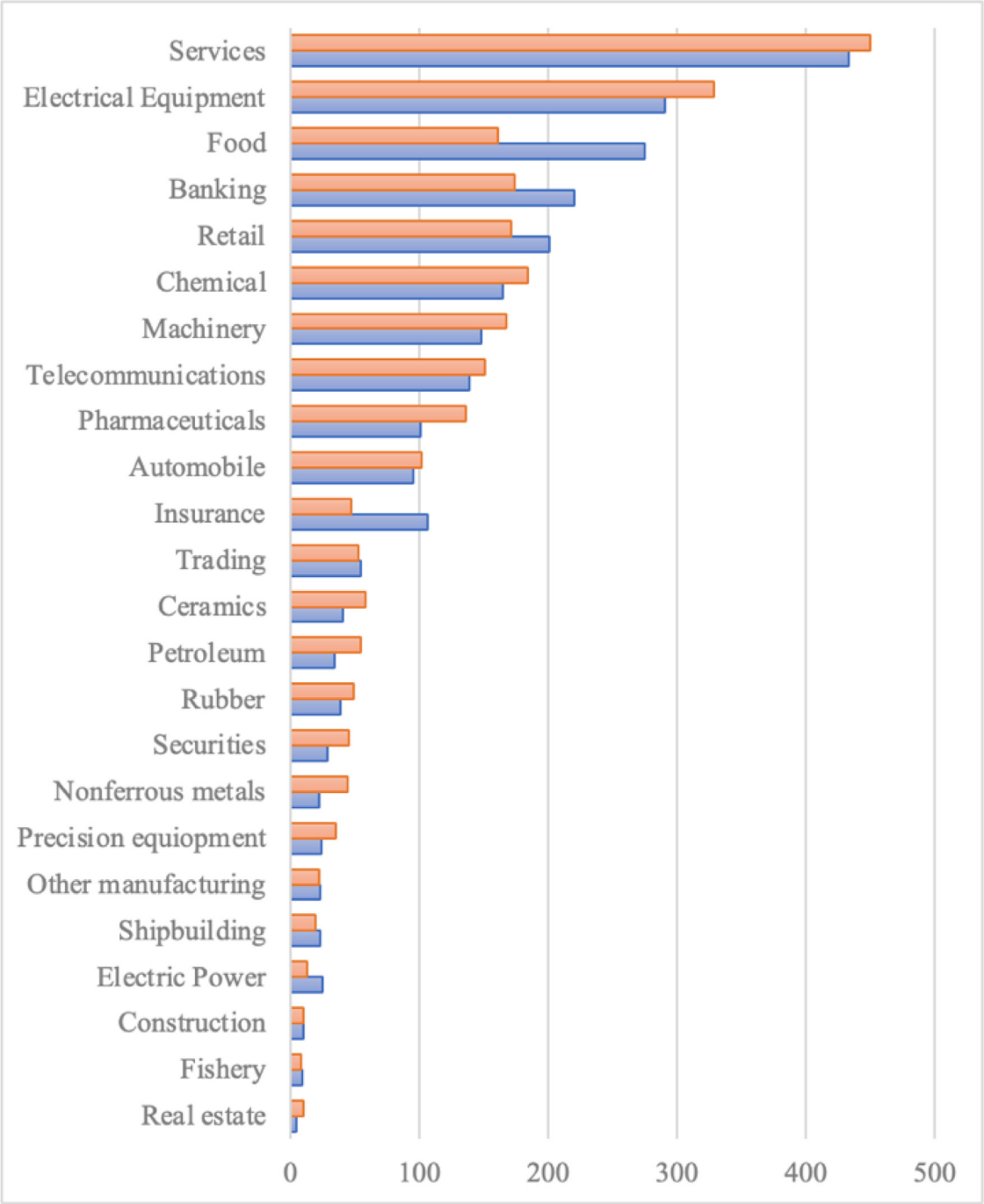

Fig. 4. Distribution of extracted the number of risk sentences (Orange bar) and non-risk sentences (Blue bar) by industry.

while the performance of strategic risk (risk A), operational risk (risk B), and business environment risk (risk C) was higher when BERT_(c) was used. When adopting the non-majority vote method, the overall accuracy might be higher if these two methods are counted separately. We also analyzed BERT_(abc) as a model integrating all of BERT_(a), BERT_(b), and BERT_(c).

Robustness check and model evaluation

To check the robustness of the annotation data, we used Fleiss' copper value (Landis & Koch, 1977), a coefficient that measures the degree of agreement between evaluations by multiple evaluators.

In the risk sentence extraction annotation, whether a sentence is a risk sentence is judged in terms of the agreement. In the risk sentence classification annotation, it is determined whether the judgments of each risk category are completely consistent. The evaluation of the copper value, based on Landis & Koch (1977), is used as a customary standard.

Risk extraction and risk classification analyses were performed by dividing the obtained annotation data into training, test, and validation data in a ratio of 7:2:1. Training data were used to build the models, and test data were used to evaluate the models. Validation data were used to evaluate the fit to unknown data.

The precision, recall, and F1 scores were used to evaluate the models. Precision represents the number of true positives divided by the sum of

Table 3
Interpretation of Fleiss's Kappa (Landis & Koch, 1977).

k	Interpretation
< 0	Poor agreement
0.01 – 0.20	Slight agreement
0.21 – 0.40	Fair agreement
0.41 – 0.60	Moderate agreement
0.61 – 0.80	Substantial agreement
0.81 – 1.00	Almost perfect agreement

Table 4
The number of sentences by annotated risk classification (Uni-labeled).

Number of Sentences	Annotated risk classification
527	Business environment risk
522	Strategic risk
507	Operational risk
443	Financial risk
235	Hazard risk

true and false positives.

$$Precision = \frac{True\ positives}{True\ positives + False\ positives}$$

Recall is the number of true positives divided by the sum of the true positives and false negatives. It represents the percentage of relevant classifications that the model successfully predicts compared to existing relevant data.

$$Recall = \frac{True\ positives}{True\ positives + False\ negatives}$$

Generally, there is a trade-off between precision and recall. The F1 score was obtained by computing the harmonic mean of precision and recall.

$$F1score = \frac{2Recall\ Precision}{Recall + Precision}$$

Results

After extracting sentences related to business risks from the securities reports published by companies that comprise the Nikkei 225, 5007 sentences were extracted as valid sentences. The data were annotated using the method described above (Step 3 in Fig. 1), through which, we found 2494 sentences that contained risk (defined as risk sentences) and 2513 sentences that did not contain risk. Distribution of extracted risk sentences (Orange bar) and non-risk sentences (Blue bar) by industry are shown in Fig. 4. Top industries in this figure; services, pharmaceuticals, chemicals, machinery, banking, retail, food, and telecommunications are the main industries in the Nikkei 225.

To check the robustness of these extraction and classification annotations, we calculated kappa values. For the risk sentence extraction annotation, a value of 0.77 was obtained. For the classification annotation, a value of 0.56 was obtained. These results were interpreted as substantial for the risk extraction annotation and moderate for the classification annotation, as shown in Table 3.

Table 4 illustrates the breakdown of each sentence classified because of the annotation. "Uni-labeled" in this table means that only one label was assigned. Similarly, "bi-labeled," as shown in Table 5, indicates a sentence with two labels assigned to it.

Table 4 shows that the most frequently annotated category was business environment risk. There was a similar amount of risk sentences classified as strategic risk. Additionally, hazard risk was the least annotated compared to the other four risk categories. Table 5 also shows the breakdown of each sentence classified by annotation, but with two labeled sentences.

Table 5
The number of sentences by annotated risk classification (Bi-labeled).

Number of Sentences	Annotated risk classification pairs
64	Operational risk, Hazard risk
47	Strategic risk, Business environment risk
45	Business environment risk, Financial risk
32	Operational risk, Business environment risk
30	Business environment risk, Hazard risk

Bi-labeling means that there are two labels for a single risk sentence; in other words, it is not possible to determine one label or the other. Table 5 suggests that operational risk and hazard risk have a relatively strong inter-relationship.

The results of the experiments are as follows: The datasets used for the evaluation are listed in Table 6.

With this training and test data, the results of the five analysis methods described in Section 3.5 of the model development are presented. The evaluation results of risk sentence extraction are presented in Table 7. Macro averages were used to calculate precision, recall rate, and F1 score.

In terms of precision, there was no significant difference in the superiority of methods, except for the expert-based method. However, the effectiveness of the expert-based method is also shown in terms of recall. The F1 score indicates that both machine learning and deep learning methods can achieve better overall results than expert-based methods.

The results of the classification performance of the risk sentences are presented in Table 8. A total of 2494 risk sentences were included in the data.

In all the machine learning methods, the precision was lower than that of Recall. This was especially true for RF, where the overall performance was lower than that of the SVM and LR. In deep learning, the relationship between precision and recall is like that in machine learning. In BERT, the performance was better than that of machine learning, but in LSTM, the performance was worse than that of machine learning.

Industry-specific risk classification analysis

In Fig. 5, we organize the categorized risk sentences from the securities reports of the companies in each industry to comprehend the risks specific to each industry.

Fig. 5 shows that the construction and electricity industries have the lowest proportion of strategic risk. Conversely, the service industry has a higher-than-average proportion of strategic risk. Hazard risk is extremely low in the insurance industry. Business environment risk accounts for at least 14% in all industries, while financial and hazard risks account for less than 5% in some industries, and risk statements related to strategic and operational risks cannot be confirmed in some industries.

Discussion

Retrieved data

From Fig. 3, in all industries, there was no extreme bias in the ratio between risk and non-risk sentences, and they could be treated as appropriate training data. The results of the five risk sentences—annotated as uni-labeled in Table 4—show that there was no difference in the total number of sentences, except for hazard risk (235 sentences). On the other hand, strategic risk (522 sentences), operational risk (507 sentences), business environment risk (527 sentences), and financial risk (443 sentences) are considered common, regardless of the industry. However, the number of E: hazard risk sentences was limited because there was a clear distinction between industries that considered it necessary to explicitly state this type of risk in securities reports and those that

Table 6
Summary of data of sentences used for risk extraction analysis.

	Positive samples(Risk sentences)	Negative samples(Non-risk sentences)	Total
Training data	1751	1754	3505
Test data	504	497	1001
Verification data	239	262	501

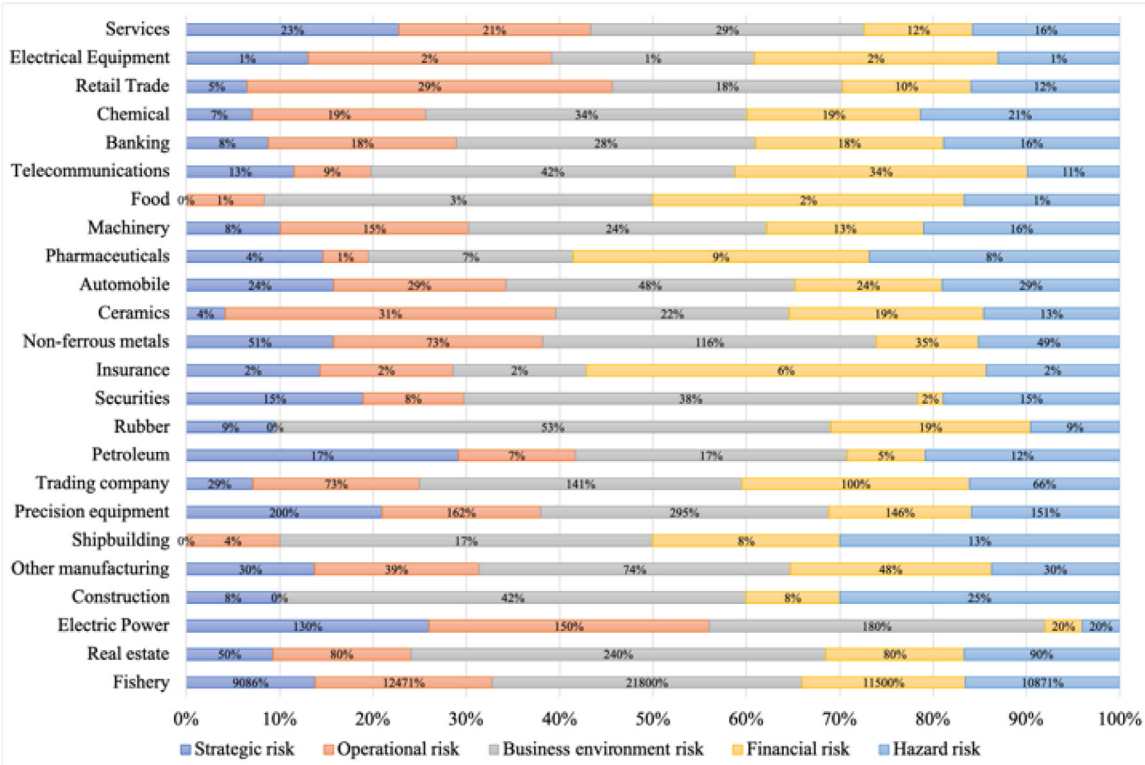

Fig. 5. Categorized risk sentences in each industry.

Table 7
Evaluation results of risk sentences extraction analysis.

Analysis types	Models	Precision	Recall	F1 score
Expert base	Hypothetical expressions	0.75	1.00	0.85
	Possible expressions	0.44	0.95	0.60
Machine-learning base	SVM	0.97	0.92	0.94
	LR	0.95	0.95	0.95
	RF	0.94	0.94	0.94
Deep-learning base	Bi-LSTM	0.94	0.94	0.94
	BERT	0.96	0.95	0.95

Table 8
Classification performance of risk sentences in each model.

Analysis types	Models	Precision	Recall	F1 score
Expert base	PM	0.41	0.40	0.40
Machine-learning base	SVM	0.58	0.95	0.72
	RL	0.57	0.97	0.71
	RF	0.36	1.00	0.52
Deep-learning base	LSTM	0.55	0.96	0.69
	BERT_(a)	0.78	0.98	0.86
	BERT_(b)	0.67	0.99	0.80
	BERT_(c)	0.72	0.98	0.83
	BERT_(abc)	**0.83**	0.97	**0.89**

did not. In this table, the most common combination of B: operational risk and E: hazard risk (64 sentences) is found in bi-labeled sentences, in Table 5.

Risk extraction analysis

As shown in Table 6, a risk sentence extraction model was run using the data. The performance results are listed in Table 7. There is a large difference in precision between PM and other models (machine learning and deep learning). In our method, SVMs, LR, and the RF algorithm outperformed PM by word form in all aspects, except for the reproduction rate. The performance of PM in word form may have declined because word forms that are supposed to express risk are often used in sentences that do not express risk. Table 9 shows some examples of sentences that were not properly extracted using the PM method.

Sentence 1 in Table 9 was extracted as a risk sentence containing a possible expression; however, the sentence explains the risk to be described later. Thus, the risk content was not extracted. Table 9: Sentence 2 was not extracted as a risk sentence, but it is a variation of the phrase “there is a risk,” and several companies used other variations as well. Sentence 3 in Table 9 was extracted as having a hypothetical expression, but in this case it was used to explain a condition and did not indicate an uncertain event. Furthermore, sentences such as Sentence 4 in Table 9 do not fall under either possible or hypothetical expressions. Rather, they express risk by using words such as “price fluctuation” (which contains the nuance that the fluctuation itself cannot be predicted) or a roundabout way of saying “there is no guarantee.” Therefore, they cannot be extracted as risk sentences. Especially, when extracting cases such as sentence 4, there may be limits to extracting risk by word form.

Table 9
Examples of sentences not properly extracted by the PM.

No.	Sentences	Corporations	Found or not by PM
1	"Risks that may affect the financial position and results of operations by our group include the following."	Fujifilm Holdings Corp.	Incorrectly extracted by PM even though it is not a risk statement.
2	"However, because of the precision of the products, there is a risk that minor shocks during the loading, unloading and transportation stages may result in total damage, and that damage may increase owing to the high price of the products."	Canon Inc.	Risk statement but could not be extracted by PM.
3	"The performance of our group tends to be better in the second half of the year when compared with the first half."	Yokohama Rubber Co., Ltd.	Incorrectly extracted by PM even though it is not a risk statement.
4	"There is no guarantee that such sharp price fluctuations will not be prolonged, or that they will not occur in markets that have not experienced such fluctuations in the past."	Suzuki Motor Corp.	Risk statement but could not be extracted by PM.

When the composition of an entire risk sentence was evaluated using SVMs, LR, and the RF algorithm, the proposed method was more effective than the other methods because the sentence was formulaic and lacked diversity. The fact that the sentences are formulaic can be partly attributed to the fact that the target of the analysis is a securities report, a document whose format is regulated by law. The effectiveness of bi-LSTM lies in its use of bidirectional information from the beginning to the end of the sentence and from the end to the beginning of the sentence to represent risk. In other words, the context or word order has a strong effect on risk expression. The optimal results from BERT can be attributed to its improved classification performance owing to its ability to fill sentences and determine the relationship between two sequences through prior learning.

Risk classification analysis

The overall evaluation results of risk sentence classification using keywords were quite low for all values. However, the overall performance improves when we focus only on each risk category individually. In other words, although the use of keywords is effective in judging the content of individual risks to some extent, the performance is still not very high, and misjudgments may be more frequent when judging sentences with multiple labels. Therefore, this method is inappropriate for determining keywords.

Regarding supervised machine learning, the performance of the SVMs, LR, and RF algorithms was considerably lower than that of risk sentence extraction. This result is particularly noticeable in the case of the RF algorithm, where the reproduction rate is lower than that of all other methods. This may be because the ratio of sentences containing each risk category to the total amount of data was smaller than that for risk sentence determination.

Instead, bi-LSTM and BERT also had high performance, albeit lower than that of risk sentence extraction, demonstrating the pure performance of supervised machine learning and deep learning. A characteristic feature of BERT is that its recall rate is much higher than its performance. This means that, without considering the context, we can assume that the words used in many risk sentences are similar, even when the risk classification is different. Besides, the high performance of BERT can be attributed to its improved classification performance because of its ability to fill in sentences and determine the relationship between two sequences through prior learning and risk sentence extraction.

Additionally, as a preliminary step of the evaluation, we used BERT to extract risk sentences using data only from the extracted sentences and experimented again. However, there was no noticeable improvement in performance despite increasing the proportion of positive examples in the data. Rather, in BERT, the performance improved when more non-risk sentences were included. This may be because the extraction reduced the amount of data used for training. However, the fact that the performance of these experiments did not change significantly suggests that the model for risk sentence classification can perform effectively even when some statements are not risk sentences.

In conclusion, the order of the words or context of the sentence is likely to be more important than the expression in risk extraction and classification.

Industry-specific risk classification analysis

Fig. 4 shows that the industries with the lowest strategic risk, construction, and electric power, are both characterized by strong state-owned businesses in Japan, and the low percentage of strategic risk seems reasonable given that they are not exposed to a strong competitive environment. Conversely, we find that strategic risk accounts for a higher average share in sectors such as services, which have a relatively high rate of opening and closing (highly competitive) relationships compared to other industries (METI, 2019). In contrast, the industry with the highest operational risk is the pharmaceutical industry (39%). This result can be attributed to the fact that risks in the workplace, including manufacturing processes, are more directly related to human lives than risks in other industries. The low hazard risk in the insurance industry is an interesting result. It is assumed that hazard risk, generally, is not considered a risk for the insurance industry because it has a positive effect on business performance. Thus, the percentage of risk classification shows how each industry perceives each risk.

As Fig. 6 shows, these results can also be used to identify trends related to individual companies. The figure shows the results of comparing the risk distribution of three companies that represent the Japanese telecommunications industry.

This figure shows that the percentage of financial risk in the total risk of Softbank is slightly higher (30%) than that of the other two companies. Accordingly, we can infer that the investment business is a large part of Softbank's overall business. According to the 2020 annual reports of NTT DOCOMO. And Softbank, the ratio of investments and other assets to total assets is approximately 8% for NTT DOCOMO and 59% for Softbank. Conversely, operational risk was not identified for KDDI, but this was because of misjudgment, and two cases that were judged as hazard risks were operational risks. In this case, the number of cases judged as a hazard risk by the KDDI is 0%. This may be because only the items related to hazard risk were itemized in the relevant securities report and were omitted from the original data in this experiment. For the KDDI, the number of original risk statements was only 11, which shows that a few misjudgments can have a large impact on the overall results. This is a limitation of the method for companies with few risk sentences in their securities reports.

Contribution to the existing literature

Analyzing unstructured corporate reports is challenging for both humans and computers. In a manual analysis, a person possessing advanced subject knowledge must scrutinize large volumes of documents; the volume and complexity of data make it cumbersome (Cohen et al., 2012). However, computerized analyses also face difficulties, such as the inability to handle irregular expressions and new risk factors (Lewis & Young, 2019).

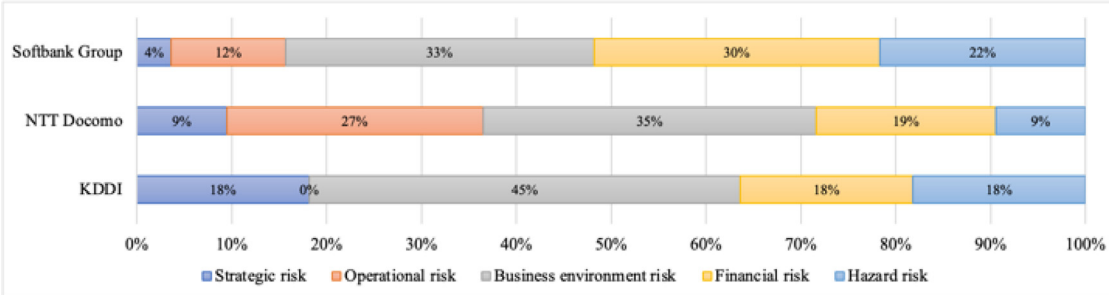

Fig. 6. Risk categories of the three companies in the Japanese telecommunications industry.

The study's academic contribution is that it proposes a framework that compensates for the shortcomings of both humans and computers in risk analysis. Extant literature on risk analysis has generally focused on one of these aspects, and it has not explored the strengths of humans and computers. This study showed that humans setting appropriate definitions and constructing sufficient supervised data could significantly contribute to the success of computerized risk extraction and classification. That is, properly defining risk factors in advance would facilitate the creation of a flexible and accurate model that requires a small amount of teacher data.

This paper discusses a scheme to seamlessly perform a series of risk extraction and classification tasks. Through a series of experiments in which the extraction and classification analyses were conducted by machines based on the process of risk definition and assessment by humans (with expertise), we demonstrated the usefulness of hybrid risk extraction and classification by both humans and machines.

Implications for practice

As the risk environment surrounding companies changes drastically, the extraction and classification of useful risk statements from ambiguous expressions and unstructured information is very costly, whether using humans or computers. With risk factors changing on a daily basis, conventional pattern matching does not allow for flexible context-aware risk analysis.

Our results demonstrate that context-aware models can be built even with limited teacher data using the proposed framework: setting appropriate definitions, constructing teacher data, extracting risk, and classifying risk. Companies and industries can use these results to identify risk trends. Furthermore, the proposed framework's accuracy shows that the framework can effectively recognize the overall risks pertaining to a given industry or company.

Our method does not necessarily require extremely sophisticated domain knowledge in the construction of supervised data, and the analytical framework can be constructed quickly and at low cost, and the results can be checked immediately. As new risk factors and unstructured information with ambiguous expressions increase, using the same model becomes unreasonable. Therefore, this study presents a framework that enables context-sensitive risk analysis using computers and definitions and supervised data created by humans.

Conclusion

As the environment surrounding companies changes drastically, there is a wide range of risk factors and a limit to the number of risk sentences that can be classified manually. To obtain risk sentences for classification, it is necessary to extract them appropriately from a large amount of textual data. Although the conventional risk extraction and classification tasks performed by humans are assumed to be accurate, they have limitations in terms of scalability. In contrast, machine analysis solves the scalability issue, but there is a lack of discussion on the risk definition to be assumed; further, the accuracy of annotated data has not been evaluated. This study discussed a scheme to seamlessly perform a series of risk extraction and classification tasks. This study examines the usefulness of hybrid risk extraction and classification by both humans and machines through a series of experiments in which extraction and classification analyses by machines are conducted based on the process of risk definition and evaluation by humans.

In this study, we defined risk, clarified the definitions of the five categories, constructed annotated data, and quantitatively evaluated the validity of the data. We used various methods to extract and classify sentences from securities reports to describe the risks assumed by companies for use in corporate valuations. Specifically, we checked the difference in accuracy between the keyword-based methods, supervised machine learning, and deep learning. The results indicated that the BERT learning method was the most effective in terms of precision, recall rate, and F1 score, whereas the method using PM in word form was the least effective in risk sentence extraction. The results also suggest that the definitions of risk and the risk classification presented in this study are valid. Also, although it is difficult to use a specific expression as a cue, the supervised machine learning method was effective in all cases where the risk sentence judgment was based on the words contained in the sentence, and the deep learning method was more effective when considering the order of the strings in the sentence. In this study, we proposed a method for identifying the risks contained in documents such as financial statements and annual reports, and our method contributes to improving the accuracy of the information for corporate value evaluation. Such an approach may be useful for investors to make profitable decisions.

Limitation and future work

We have various aspects to be addressed in future studies. The first is the versatility of datasets. In this study, we used securities reports for analysis because of their ease of accessibility and uniformity of format; however, other documents contain risk-related content as well. For example, integrated and analyst reports contain risk statements. Therefore, it is necessary to confirm whether the same results can be obtained in other studies. Second, the extracted and classified risks are not independent entities. Risks do not occur in isolation—except in the case of natural disasters—and are often caused by the occurrence of various events (risk factors), including other risks. We define this phenomenon in terms of the causal relationships among the risk factors. Furthermore, one risk factor may give rise to multiple other risk factors. For example, a risk factor, such as soaring raw material costs, may be caused by another risk factor, even within the same company, such as natural disasters or changes in international conditions. To understand and assess the risks that companies face, it is necessary to understand not only the risk factors but also their ultimate effect on business performance. That is, it is important to create a hierarchy of risk factors when conducting a risk analysis, and this issue needs to be addressed in the future. An causal search method might be used for this purpose (Izumi & Sakaji, 2019). Furthermore, this study did not mention the time-series. Even for the

same type of industry or company, it is assumed that the type of risk differs depending on the time-series. Confirming the time-series analysis using the results of this study will lead to a better understanding of risk trends by industry. Furthermore, only the securities reports issued by Japanese companies were discussed. This study presents concrete experimental results that show that automatic risk extraction and classification are possible on a large-scale using appropriate risk definitions. Based on the ideas of this study, it would be meaningful to conduct experiments in other languages. In the case of actual business use, it is also important to consider the lower limit of the practical accuracy of risk extraction and classification. The focus of this study was not to sufficiently discuss the lower limit of accuracy. This is an important direction for future research.

The results indicated that diversified risks can be extracted using this method to enable early risk management. Additionally, the fact that these definitions and methods were also effective for classification and analysis suggests that it may be easier to grasp the outlines of newly-identified risks. Furthermore, we confirmed that it is possible to distinguish risks by context rather than by keywords, as has been found in previous studies. This approach is expected to enable us to extract and classify more risks from a wide variety of data sources rather than focusing merely on specific data sources.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jjimei.2022.100096.

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